

# HOUJIANG LIU

Ph.D. Student, School of Information, The University of Texas at Austin

## CONTACT

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**E-mail:** liu.ho at utexas dot edu

**Website:** <https://jialingjia.github.io/houjiang-website/>

**Google Scholar:** <https://scholar.google.com/citations?user=a8xtohlAAAAJ&hl=en>

**Office:** 1616 Guadalupe St, UTA 5.520, Austin, TX, 78701, USA

## EDUCATION

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**University of Texas at Austin, School of Information, 2021–present**

Ph.D. Student of Information Studies/Science

*From Automation to Agency: Evaluating Computational Support for Human Research Ideation*

Committee: Matthew Lease (advisor), Jacek Gwizdka, Ahmer Arif (Barnard College), & Aniket Kittur (CMU)

**Northeastern University, College of Arts, Media and Design, June 2020**

M.F.A. in Design

*Enriching Gifts, Experience Design to Understand Human Relationships*

Committee: Miso Kim (advisor), Kristian Kloeckl, & Meryl Alper

**Jilin University, College of Physics, July 2015**

B.S. in Applied Physics

## RESEARCH AREAS

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My research in **Human-centered Artificial Intelligence** centers on **Human-computer Interaction** and **Design Research**, with a particular focus on integrating human-centered principles into AI system design.

A major strand of my PhD research involves collaborating with key stakeholders in the online information ecosystem, such as journalists, fact-checkers, and the public, **to design responsible AI to protect information integrity** [P11, P10, P9, M2]. More recently, I have expanded my interests to AI-assisted scientific discovery, **examining how large language models (LLMs) and LLM-based agents shape different stages of research activities** [P12, E4, E3, M4].

Building on my earlier MFA research, I continue to investigate **older adults' tech literacy and health** [P8, P7, P3, P2, P1, M1] and **information visualization** [P6, P4, D1].

## AWARDS

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**Best Paper Honorable Mention**, ACM CSCW, 2024

Recognition is awarded to the top 4% of submitted papers. [[Official announcement](#)]

**Information is Beautiful Awards (longlist)**, Data Visualization Society, 2022

*The Information is Beautiful Awards (longlist) highlight exceptional global entries in data visualization, infographics, and information art, serving as showcases for finalists each year.* [[Showcase](#), [Software](#), [Code](#)]

**First Design Award**, Health x Design Challenge, Mad\*Pow, 2019

Winner of the 2019 Health Experience Design Challenge, recognizes innovative projects that promote health, financial wellbeing, and social impact, organized by Mad\*Pow (now part of BORN Consulting Group) and Robert Wood Johnson Foundation. [[Press](#)]

**Innovative Design Award**, Reinventing Tencent Map, Tencent, 2014

Winner of the 2014 Tencent Map Redesign, recognized for pioneering the integration of Augmented Reality (AR) features into the mobile application.

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## RESEARCH GRANTS & SCHOLARSHIP

**Research Grant Award (Track: AI and Human Autonomy)**, Cosmos Institute, 2026

**E. Glynn Harmon Excellence Fund**, University of Texas at Austin, 2026

**Outstanding Graduate Research Fellowship**, University of Texas at Austin, 2025-26

**Professional Development Award**, University of Texas at Austin, 2023, 24, 25

**Summer Research Grant**, University of Texas at Austin, 2023

**Deans' Award for Research Scholarship and Creativity**, Northeastern University, 2020

**Summer Research Grant**, Northeastern University, 2019

**Graduate Student Scholarship**, Northeastern University, 2018–20

**Undergrad Merit Scholarship** (plus awarded direct admission to graduate program), Jilin University, 2012–13, 2015

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## PUBLICATIONS (Authors with equal contributions are marked with \*)

### Peer-Reviewed Conference Proceedings and Journal [P]

[P12] Sanjana Gautam, **Houjiang Liu**, Yujin Choi, and Matthew Lease. How Researchers Navigate Accountability, Transparency, and Trust When Using AI Tools in Early-Stage Research: A Think-Aloud Study. In: *Proceedings of the 9th ACM Conference on Fairness, Accountability, and Transparency*, pp. 1494–1512, 2026, Montréal, Canada, 2026. [[DOI](#), [arXiv](#)]

[P11] **Houjiang Liu**, Jacek Gwizdka, and Matthew Lease. Exploring Multidimensional Checkworthiness: Designing AI-assisted Claim Prioritization for Human Fact-checkers. *Proceedings of ACM Human-Computer Interaction*, 9 (7): CSCW292, 49 pages, 2025. [[DOI](#), [arXiv](#), [Code](#), [Slides](#), [Video](#)]

[P10] **Houjiang Liu\***, Anubrata Das\*, Alexander Boltz\*, Didi Zhou, Daisy Pinaroc, Matthew Lease, and Min Kyung Lee. Human-centered NLP Fact-checking: Co-Designing with Fact-checkers using Matchmaking for AI. *Proceedings of ACM Human-Computer Interaction*. 8, CSCW2, Article 423, 44 pages, 2024. [[DOI](#), [arXiv](#), [Slides](#)] **Best Paper Honorable Mention (top 4%)**

[P9] Anubrata Das, **Houjiang Liu**, Venelin Kovatchev, and Matthew Lease. The state of human-centered NLP technology for fact-checking. *Information Processing & Management*, 60(2) :103219, 2023. [[DOI](#), [arXiv](#)]

[P8] **Houjiang Liu\***, Estefania Ciliotta Chehade\*, William Albert, and Miso Kim. An Empirical Study of How Service Designers Use Metrics. *She Ji: The Journal of Design, Economics, and Innovation*, 9(3) :402–427, 2023. [[DOI](#), [Blog](#)]

- [P7] Mingzhu Li, Miso Kim, Liqing Huang, and **Houjiang Liu**. Designing for people you can't meet: Persuasive design to nurture seniors' participation in community services during the COVID-19 pandemic. *The Design Journal*, 26(6) :934–955, 2023. [\[DOI\]](#)
- [P6] Sara Colombo, Estefania Ciliotta Chehade, Lucia Marengo, **Houjiang Liu**, Piero Molino, and Paolo Ciuccarelli. Design for Emergency: How Digital Technologies Enabled an Open Design Platform to Respond to COVID-19. *Interacting with Computers*, 35(5) :650–665, 2023. [\[DOI\]](#), [Website](#), [Blog](#), [Video](#)
- [P5] **Houjiang Liu** and Miso Kim. Is gift a token of gratitude or an imposition of identity? Facilitating positive consequences of gift-giving with receiver-centred design. In: *Proceedings of the 10<sup>th</sup> International Association of Societies of Design Research (IASDR)*, Milan, Italy, 2023. [\[DOI\]](#), [Slides](#)
- [P4] Jiabao Li, **Houjiang Liu**, Jilie Zeng, Di Wu, Ying Ding, and Alec McGail. Shifting Winds: Gendered Structures of Academic Mentorship. In: *Proceedings of 2022 IEEE VIS Arts Program (VISAP)*, pp. 14–29, ieeexplore, Oklahoma City, USA, 2022. [\[DOI\]](#), [Website](#), [Code](#)
- [P3] **Houjiang Liu**, Ray Lc, Miso Kim, Concettina Cormio, and Meixin Yu. Designing for Distance Nursing: Reconnecting Nursing Students with Senior Home Residents During COVID-19. In: *Proceedings of the 9<sup>th</sup> International Association of Societies of Design Research (IASDR)*, Hong Kong, 2022. [\[DOI\]](#), [Slides](#)
- [P2] **Houjiang Liu**, Miso Kim, Mingzhu Li, and Shruthi Lakshmi Narayan. Technology, Autonomy, and Participation: Designing Community Games and Services to Enhance Older Adults' Technology Literacy. In: *Proceedings of the 9<sup>th</sup> International Association of Societies of Design Research (IASDR)*, Hong Kong, 2022. [\[DOI\]](#), [Slides](#)
- [P1] **Houjiang Liu**, Miso Kim, Canqun He, and Tia Thomson. Autonomy as a Design Principle: Service Design for the Technology Literacy of Older Adults. In: *Proceedings of the Cumulus Conference on Design Culture(s)*, Rome, Italy, 2021. [\[DOI\]](#)

#### Lightly Reviewed Extended Abstracts, Short Papers, & Posters [E]

- [E4] Huimin Xu, **Houjiang Liu**, Yan Leng, and Ying Ding. Adapting to LLMs: How Insiders and Outsiders Reshape Scientific Knowledge Production. In: *International Conference on the Science of Science and Innovation*, Boulder, USA, 2026. [\[arXiv\]](#)
- [E3] **Houjiang Liu\***, Yujin Choi\*, Sanjana Gautam, Gabriel Jaffe, Soo Young Rieh, and Matthew Lease. Effects of Varying Control vs. Automation in Human-in-the-loop Research Ideation. In: *Amazon Research Day, Workshop on AI co-scientist: Accelerating Scientific Discovery through Intelligent AI Collaboration*, Palo Alto, USA, 2026. [\[pdf\]](#)
- [E2] Anubrata Das, **Houjiang Liu**, Venelin Kovatchev, and Matthew Lease. The Need for Human-centered Design in Fact-checking Research. In: *Proceedings of the 1st Information Processing & Management (IP&M) Conference*, Xiamen, China, 2022. [\[pdf\]](#)

- [E1] **Houjiang Liu**, Enriching Family Gifts: Reciprocity in Family Relationships (Extended Abstract of C5). *NA PhD by Design Symposium*, New York University, USA, 2020. (Symposium canceled due to COVID19 shutdown)

#### Pre-print Manuscripts [M]

- [M4] **Houjiang Liu\***, Yujin Choi\*, Sanjana Gautam, Gabriel Jaffe, Soo Young Rieh, and Matthew Lease. Who Owns Creativity and Who Does the Work? Trade-offs in LLM-Supported Research Ideation. 2026. [[arXiv](#)]
- [M3] **Houjiang Liu**, Yiheng Su, and Matthew Lease. Rhetorical XAI: Explaining AI's Benefits as well as its Use via Rhetorical Design. 2025. [[arXiv](#)]
- [M2] **Houjiang Liu\***, Li Shi\*, Yian Wong, Utkarsh Mujumdar, Dan Zhang, Jacek Gwizdka, and Matthew Lease. Reducing Confirmation Bias on Controversial Issues through LLM-Generated Multi-Persona Debates. 2024. [[arXiv](#)]
- [M1] Jiaxin An, Siqu Yi, Yao Lyu, **Houjiang Liu**, and Yan Zhang. Conversational Agents for Older Adults' Health: A Systematic Literature Review. 2025. [[arXiv](#)]

#### DESIGN EXHIBITION

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- [D4] Yinan Dong, Oyinkan Ifidon-Ola, **Houjiang Liu**, Kee Moon Jang, Deniz Aydemir, Fábio Duarte, Carlo Ratti. Biophilia: Mapping the Emotional Pulse of Urban Nature. In: *2025 Information+ Conference*, Boston, 11/14-16. [[Exhibition](#)]
- [D3] Jiabao Li, **Houjiang Liu**, Jilie Zeng, Di Wu, Ying Ding, and Alec McGail. Shifting Winds: Gendered Structures of Academic Mentorship. In: *2022 IEEE VIS Arts Program (VISAP)*, Oklahoma City, 10/16-21. [[Exhibition](#)]
- [D2] Sara Lenzi, Paolo Ciuccarelli, **Houjiang Liu**, and Yuan Hua. Data Sonification Archive. In: *2022 Design Research Week*, Boston, 4/25-29 [[Website](#)]
- [D1] **Houjiang Liu**, Yingyin Xu, Anni Xu, How Food is Served in Early Care Center. In: *Northeastern University Visualization Consortium (NUVis) Demofest*. Boston, 2019.

#### INVITED RESEARCH TALKS (Referencing previous academic work)

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- [M5] Assessing Creativity vs. Efficiency Tradeoffs across Control Levels of LLM-Supported Research Ideation
- *CosmicAI Biweekly Seminar*, February 2026.
- [P11] Exploring Multidimensional Checkworthiness: Designing AI-assisted Claim Prioritization for Human Fact-checkers.
- *iSchools Global Doctoral Seminar Series*, August 2025.
- [M2] Generating Multi-Persona Debates with LLMs to Explore Diverse Perspectives on Controversial Topics.
- *Kungfu.AI* guest lecture, May 2025.
  - *Good Systems Annual Symposium, Shaping the Future of Ethical AI*. University of Texas at Austin, Spring 2025.

- *iSchool AI Showcase: Community-Engaged Research on AI and Society*. University of Texas at Austin, Fall 2024.

**[P8]** An Exploration of Service Design Metrics.

- *Research, Innovation, Scholarship, Entrepreneurship (RISE), Virtual EXPO*. Northeastern University, 2021.

**[P2,3]** Participatory Design for Senior Healthcare: Service Design to Improve Seniors' Health Literacy and Autonomy.

- *Research, Innovation, Scholarship, Entrepreneurship (RISE)*. Northeastern University, 2020.

## PROFESSIONAL EXPERIENCE

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### Academic

Graduate Research Assistant, AI & Human-Centered Computing Group, University of Texas at Austin. 2021–present

*Conduct interdisciplinary research on human-AI interaction, fields include creativity support and fact-checking. Lead research activities across funded projects, including study design, prototyping, data analysis, publication, and project coordination.*

Design Researcher, Center for Design, Northeastern University. 2020–21

*Conducted design research on service design and data visualization. Applied qualitative and design methods to investigate older adult needs of technology.*

Lecturer, College of Arts, Media and Design, Northeastern University. 2020–21

*Developed and taught courses in interaction design and user experience. Mentored students on design research, prototyping, and human-centered design practices.*

### Professional

User Experience Designer, SeetaTech, Beijing. 2017–18

*Led user experience design for AI-powered products, translating user insights into product strategies, interaction flows, and interface designs.*

Interaction Designer, JD.com, Beijing. 2015–2017

*Designed user experiences for e-commerce platforms and services across web and mobile products. Conducted user research, created interaction frameworks, and worked closely with product and engineering teams to deliver customer-centered solutions.*

Interaction Design Intern (2015 Spring, JD.com; 2013 Summer, Lenovo; 2012 Summer, Midea Group)

## SERVICE

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### Peer Reviewer (5 x Recognized for Outstanding Reviews)

- *International Journal of Human-Computer Interaction* 2026
- *International Journal of Intercultural Relations* 2026
- *Information Processing & Management* 2026
- *ACM SIGCHI Conference on Human Factors in Computing Systems (CHI)* 2026 (1 outstanding review)

- *ACM SIGCHI Conference on Collective Intelligence (CI) 2025*
- *ACM SIGCHI Conference on Designing Interactive Systems (DIS) 2025*
- *ACM SIGCHI Conference on Computer-Supported Cooperative Work & Social Computing (CSCW) 2022–27 (3 outstanding review)*
- *ACM SIGCHI Conference on User Interface Software and Technology (UIST) 2024, 2026 (1 outstanding review)*
- *Design Research Society Conference (DRS) 2024*

#### **Student Volunteer**

- *ACM SIGCHI Conference on Computer-Supported Cooperative Work & Social Computing (CSCW) 2024, 2025*
- *ACM International World Wide Web Conference (WebConf) 2023*
- *ACM SIGIR Conference on Human Information Interaction Retrieval (CHIIR) 2022*
- *User Experience Professionals Association Conference (UXPA) International 2015*

#### **Ph.D. Representative, School of Information, University of Texas at Austin**

- Research Committee 2024–26
- Master Studies Committee 2021–22

#### **TEACHING EXPERIENCE**

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##### **Teaching Fellow, College of Arts, Media and Design, Northeastern University**

ARTG2400 Interaction Design, Spring 2021

ARTG2401 Interaction Design Tools, Fall 2019, Spring 2020

##### **Teaching Assistant, College of Arts, Media and Design, Northeastern University**

ARTG6330 Info Design Mapping Strategies, Fall 2020

ARTG6900 Design for Dignity, Fall 2019